

HDOC DAY_30

Problem

Write a C program to perform the following menu-driven operations on a given array:

1. Sort the elements using Bubble Sort algorithm and perform a Binary search.
2. Perform Insertion and Deletion in a given valid position.
3. Rotate the array elements to the right by k positions.
4. Exit

code

```
#include <stdio.h>

int main()
{
    int ch;

    printf("choice 1 is to sort array using bubble sort and the find element using binary search\n");

    printf("choice 2 is to perform an insertion and deletion operation at desired location in an array\n");

    printf("choice 3 Rotate the array elements to the right by k positions.\n");

    printf("exit is choice 4\n");

    printf("enter your choice\n");

    scanf("%d",&ch);

    switch (ch)
    {
        case 1:
        {
            int n;

            printf("enter the size of array\n");
```

```

scanf("%d",&n);

int arr[n];

printf("enter the elements of the array(elements should be positive integers and 2
elements should not be same)\n");

for(int i=0;i<n;i++)
{
    scanf("%d",&arr[i]);
}

for(int i=0;i<n-1;i++)
{
    for(int j=0;j<n-i-1;j++)
    {
        if(arr[j]>arr[j+1])
        {
            int temp=arr[j];
            arr[j]=arr[j+1];
            arr[j+1]=temp;
        }
    }
}

printf("the sorted array is\n");

for(int i=0;i<n;i++)
{
    printf("%d\t",arr[i]);
}

int num;

printf("\nEnter the number to be searched\n");

scanf("%d",&num);

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int ll=0;

int ul=n-1;

for(int i=0;i<n;i++)
{ int mid=(ll+ul)/2;

  if(num==arr[mid])

  {

    printf("element found at pos %d",mid+1);

    return 0;

  }

  else if(num>arr[mid])

  {

    ll=mid+1;

  }

  else{

    ul=mid-1;

  }

}

printf("element not found\n");

break;

}

case 2:

{

  int n;

  printf("enter the size of array\n");

  scanf("%d",&n);

  int arr[n];

  for(int i=0;i<n;i++)

  {

```

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        scanf("%d",&arr[i]);
    }
    for(int i=0;i<n;i++)
    {
        printf("%d\t",arr[i]);
    }
    int num,ind;
    printf("\nEnter the new element and index of the element you want to replace\n");
    scanf("%d%d",&num,&ind);
    int index=ind-1;
    arr[index]=num;
    printf("The new array is\n");
    for(int i=0;i<n;i++)
    {
        printf("%d\t",arr[i]);
    }
    break;
}
case 3:
{
    int n;
    printf("Enter the size of array\n");
    scanf("%d",&n);
    int arr1[n];
    printf("Array elements input\n");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr1[i]);
    }
}

```

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    }

    printf("original array\n");
    for(int i=0;i<n;i++)
    {
        printf("%d\t",arr1[i]);
    }

    int k;

    printf("\nenter the number by which you want to shift the array elements towards
their right\n");

    scanf("%d",&k);

    int arr2[n];

    for(int i=0;i<n;i++)
    {
        if(i<k)
        {
            arr2[i]=arr1[n+i-k];
        }

        else if(i>=k)
            arr2[i]=arr1[i-k];
    }

    printf("shifted array\n");

    for(int i=0;i<n;i++)
    {
        printf("%d\t",arr2[i]);
    }

    break;
}

```

case 4:

```
{  
    printf("exit");  
    return 0;  
}  
default:  
{  
    printf("invalid choice\n");  
    break;  
}  
}  
return 0;  
}
```